A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 21

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 6 points below. **1** Cycling of materials (like biogeochemical cycles) - recycling, re-use **1** Energy from renewable sources **1** Human activities should support ecosystems / conserve biodiversity **1** Separation of technical and biological materials – connected systems where the waste of one process feeds another (industrial symbiosis) **1** Diverse systems are more resistant to change **1** Design of products for extended use and end-of-life re-use – optimum rather than maximum production – technologies that improve efficiency and enable dismantling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new • **biodiversity** → the variety of life: habitat, species and genetic diversity • **symbiosis** → a close relationship in which both species benefit **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	6

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae **1** Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them **1** A common approach gives a level playing field, coordinated monitoring and support for less successful countries **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **abiotic** → non-living / physical factors	1
02.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Deforestation **1** Ploughing of grassland / expansion of farmland **1** Reservoir creation / flooding **1** Mineral extraction, especially open-cast mining **1** Urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Warming melts ice shelves that hold back glaciers and lubricates ice flow so land ice reaches the sea faster (Antarctic Peninsula most affected) – sea ice loss removes krill habitat and lowers albedo **1** Raised UV harms planktonic algae, krill and fish / sea urchin larvae in surface waters (land animals protected by fur / feathers) **1** Disturbance of wildlife, introduction of non-indigenous species and pathogens, fuel and sewage pollution – visitor numbers rising rapidly at a few landing sites **1** Reserves of coal, oil, gold, uranium, platinum etc – pressure to exploit will rise as reserves elsewhere deplete though currently banned and operations extremely difficult **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
03.2	**1 mark each** 1 mark for each of the 6 points below. **1** Treaty signed by 52 countries including all active there – no military activity (except supporting research), no nuclear explosions or radioactive waste dumping **1** No commercial mineral exploitation **1 Fisheries regulated by CCAMLR:** krill and some fish exploited, fisheries closed if over-exploited – no commercial exploitation of mammals **1 Tourism control:** no accommodation on land (wastes stay on ships), one ship per site with maximum 100 ashore, trained guides, keep 5 m from wildlife, no litter, cleaned protective clothing to prevent introductions **1 Waste management:** all wastes removed (fuel containers, batteries, plastics, oil) – sewage only discharged where sea currents disperse it – old waste removed unless removal causes more damage **1** Only long-term residents are research scientists and support staff – strict regulation of research impacts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	6
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 69-71; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (240 000 - 2 000) / 240 000 x 100 = 99.2% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
04.2	**1 mark each** 1 mark for each of the 3 points below. **1 Whales are k-selected:** slow growing, late maturing, one calf every few years, so populations recover very slowly **1** Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades **1** No regulation / open access to the ocean – whales were valuable (oil, spermaceti) and easily found **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Moratorium on commercial whaling since 1986 until populations recover **1** Total protection for certain species – whale sanctuaries **1** Limits on numbers and sizes taken – protection of suckling mothers and calves **1** Research into whale biology and threats (ship strikes, net entanglement) – exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since **1** Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses – whaling involves few nations and a single treaty body **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	
05.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** They are driven by renewable, low energy-density resources, especially solar energy, which will not be depleted: the hydrological cycle, the carbon and nitrogen cycles, atmospheric and thermohaline circulation **1** Most natural processes occur at low temperatures, because enzymes lower the activation energy of the reactions, so little energy is needed **1** Processes link into cycles in which the waste product of one is the raw material for the next, so raw materials are not depleted and wastes do not build up **1** A small number of abundant, simple elements are built into monomers and then into a wide range of polymers, so supplies of the raw materials are secure **1** Wastes are biodegradable and break down to non-toxic products, so they do not accumulate or poison the system **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **thermohaline** → the ocean conveyor driven by temperature and salinity • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals are dispersed after use - spread thinly through products all over the world, or lost to corrosion - so there is no concentrated deposit left to recover **1** Products are made of mixtures of materials that are difficult to separate, so recovery and re-use are expensive and often incomplete **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
05.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Design for extended use:** a product with a long life, which can be repaired, gives fewer opportunities for its materials to be lost **1 Design for end-of-life re-use:** materials that can be identified and separated easily are far more likely to be recycled **1** Separation of technical and biological materials, because metals and plastics need different reprocessing from biodegradable materials, which can be composted or digested **1** Energy from renewable sources at every stage, as in natural systems **1** Optimum rather than maximum production, so that the system supports the processes it relies on instead of maximising one output **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3

Do not write outside the box

Question	Marking guidance	Marks
05.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Sustainability depends on the total impact, which is the number of people multiplied by the impact of each person, so the same activity crosses the threshold as either rises **1** A natural process can absorb or replace what is taken only up to a rate – beyond that the resource is depleted or the waste accumulates faster than it can be broken down **1 Example:** slash and burn agriculture is sustainable where the population is small enough for each clearing to be left fallow long enough to recover, but not when land has to be re-cleared before the soil fertility has returned **1 Example:** discharging untreated sewage into a river is harmless where decomposers can break it down, but causes deoxygenation and eutrophication once the load exceeds that rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **threshold** → the concentration below which no harm is detected • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
05.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Urban populations are supplied by resources brought in from elsewhere, so the damage caused by extraction happens out of sight and often in another country **1** In a sedentary rural community the resource is local, so the problems of over-exploitation become obvious quickly as supplies dwindle, and a cultural belief in limiting exploitation to needs develops **1 Consequence:** when one area is exhausted, exploitation simply moves to another area and the consumer sees no change, so the total damage keeps growing - this has happened with timber, energy, metal minerals and fish **1 Consequence:** there is no pressure from consumers to change, so unsustainable practices continue until the resource fails on a large scale **1 Consequence:** the costs fall on the people and ecosystems where extraction happens, not on those who benefit, so the market price never reflects the true environmental cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-377; AQA spec 3.6.2-3.6.3.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Negative feedback responds to any change by re-establishing the previous equilibrium **1** Only large unusual events overwhelm the feedback so a new equilibrium forms **1 Example:** methane hydrate release at the end of the Permian causing climate change and mass extinction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **negative feedback** → a change that cancels itself out / a self-correcting cycle	3
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** In a diverse system each component is relatively less important so the loss / change of one has a small effect – alternative food sources and relationships buffer the change **1 Natural:** coral reefs and tropical rainforests with complex food webs are stable – simple polar / desert systems fluctuate **1 Human:** agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather **1 Diverse technical systems:** a society using a range of renewable energy resources is more secure than one relying on a single source **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1** Melting permafrost releasing methane **1** Reduced albedo as ice / snow melt **1** Methane hydrate release **1** Forest and peat fires **1** Faster soil decomposition **1** Ocean acidification reducing sequestration **1** Cirrus cloud formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession **1** Chalk grassland was produced by centuries of sheep grazing – without it succession to scrub and woodland resumes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 4 points below. **1** Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out **1** Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles **1** Dung creates small nutrient-rich patches supporting different plants and invertebrates – sheep eat short grasses and some bushes **1** Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting – burning removes vegetation but also kills invertebrates and allows scrub regrowth – abandoned plot dominated by scrub (55%) so few species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sheep eat short grass (some breeds eat scrub) – ponies avoid flowers but eat bracken – cattle remove long rough grass **1** Choice depends on the vegetation to be controlled and the plants to be protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground **1** Heat stimulates dormant heather seeds to germinate so young heather re-establishes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Σn(n - 1) = (40 x 39) + (30 x 29) + (20 x 19) + (10 x 9) = 1560 + 870 + 380 + 90 = 2900 **1** N(N - 1) = 100 x 99 = 9900 **1** D = 9900 / 2900 = 3.41 (accept 3.4) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B **1** The index does not identify which species are present / a rare or important species may be missed – depends on identification skill and sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
09.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1

Do not write outside the box

Question	Marking guidance	Marks
09.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Harvesting no faster than the MSY - resource never depleted **1** Mixed species plantations - more food choices, relationships and niches **1** Indigenous species - support indigenous insects, birds **1** Mixed age structure - varied light, temperature, wind so more species – continuous cover and yield **1** Selective logging - small clearings, less disturbance, long-term timber benefit **1** Retention of dead wood, buffer zones, corridors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
10.1	**1 mark each** 1 mark for each of the 2 points below. **1** Plants produce chemicals (often alkaloids) that are toxic to the herbivores that would eat them **1** In carefully controlled amounts these chemicals have beneficial physiological effects in humans **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Only a very small proportion of species have been studied for medicinal substances, so it is not known which species will be useful **1** Conserving the habitat protects all the species in it, including those not yet studied and the species they depend on **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marsupials (kangaroo / wallaby): young develop in the pouch so are easier to study than a human foetus - developmental problems **1 Hippopotamus:** hipposudoric acid skin secretion (sunscreen / antimicrobial) - burns treatment **1 Marine sponges:** proteins preventing graft rejection - organ transplants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here.	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Resources:** wood, fibres, oils, fuels, food – new food species adapted to local conditions (morama bean, yeheb, perennial maize, Kernza, bison, eland) • **Biomimetics:** whale flipper tubercles, shark skin, termite-mound ventilation, bird bones, gecko adhesion, burrs / Velcro, lotus self-cleaning, spider silk • Medicines and physiological research – genetic resources - crop wild relatives, Vavilov centres, gene pools, inbreeding • **Ecosystem services:** atmospheric composition, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination, seed dispersal, habitat provision • **Evaluation:** services go unnoticed because of dynamic equilibrium – unknown future value – economic vs moral / intrinsic arguments – conservation also needed for aesthetic and cultural reasons • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **biodiversity** → the variety of life: habitat, species and genetic diversity	9
	Model answer - Level 3 (9 marks)	
	Other species supply resources we use directly: wood, fibres such as cotton and wool, oils, fuels and food. Because most farmed species were domesticated long ago and are often kept where they are poorly adapted, wild species such as the drought-resistant morama bean, the yeheb tree that grows on poor soils, perennial maize that needs no ploughing, and bison that graze grassland too poor for cattle offer new, better-adapted food sources. Wild species also carry genetic resources: crop wild relatives in Vavilov centres hold genes for disease resistance and salt tolerance that inbred commercial varieties lack.	

Do not write outside the box

Question	Marking guidance	Marks
	Biomimetics copies adaptations into technology - humpback flipper tubercles in turbine blades, shark skin in friction-reducing coatings, termite-mound convection in naturally ventilated buildings, hollow bird bones in lightweight roofs. Medicines such as taxol, quinine, AZT and diosgenin came from wild species, and physiological research on marsupials, hippo skin and sponge proteins is helping with foetal development, burns and transplant rejection. Since only a tiny fraction of species has been studied, the unknown future value argues for conserving whole habitats.	
	Beyond resources, other species provide services we could not replace: regulation of atmospheric gases, evapotranspiration driving rainfall, the carbon, nitrogen and phosphorus cycles run by microbes, soil formation by decomposers, and pollination, seed dispersal and habitat provision. These are unnoticed precisely because they are in dynamic equilibrium, so their economic value is underestimated. On balance the resource and service arguments alone justify conservation, but they are strongest when combined with the intrinsic and aesthetic case, because many species with no known use still sustain the systems we depend on.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important because we get lots of things from other species. We get wood, cotton and food from plants and animals. Some medicines come from plants, like aspirin from willow trees. Animals can help design things, like Velcro was copied from seeds. Other species also do things for us like bees pollinating crops. If we lose species we might lose things that could be useful in the future. So we should conserve biodiversity.	
	What makes the difference	
	• Both answers cover resources, medicines, biomimetics and services, but Level 3 develops each with specific, accurate detail (which species, what property, why it matters).	
	• The question asks 'to what extent'. Level 3 weighs resource arguments against intrinsic value and reaches a reasoned conclusion; Level 1 asserts 'so we should conserve'.	
	• Level 3 explains *why* services are undervalued (dynamic equilibrium) - an explanation, not just a list.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Energy flow:** autotrophs and heterotrophs – losses by reflection, transmission, respiration – 100 000 to 1 000 units – few food chains beyond four levels – plant food most productive – grazing animals where crops impossible – pigs on wastes • **Energetics:** intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • **Control of genetics:** asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • **GM:** Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape – advantages – risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity)	25
	• **Sustainability links:** energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Sustainability definitions – population and per capita impact – attitudes of rural and urban societies • **Negative feedback:** cloud cover and albedo, CO_2 and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • **Tipping points:** rare under negative feedback – large events (Permian methane hydrate) – self-sustaining change from human activities • **Diversity and resistance:** reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides) – links to climate change chapter – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	

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